

[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

Dec-25-0494

MCA-6104 (Discrete Mathematical Structures)

MCA 1st (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from section A, B, C and D. Question no.9 is compulsory.

SECTION - A

1. Define with example proposition and predicate logic. (10)
2. (a) Define Cartesian product of sets and explain with example. (5)
(b) Prove that the function $f(x) = x^2$ is not one-one but onto from $\mathbb{R}$ to $[0, \infty)$ (5)

SECTION - B

3. (a) In how many ways can the letters of the word MATHEMATICS be arranged? (5)
(b) State and prove the pigeonhole principle with an example. (5)
4. (a) Solve the recurrence relation :
 $T(n) = 2T(n-1) + 3, T(1) = 1$ (5)
(b) Find the probability of getting at least one head when 3 coins are tossed simultaneously. (5)

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SECTION - C

5. (a) Define Eulerian and Hamiltonian graph with suitable example. (5)
(b) Prove that a connected Graph has an Eulerian circuit if and only if every vertex has even degree? (5)
6. (a) Draw a binary tree for the expression : $(A+B) * (C-D)$. (5)
(b) Explain Prim's algorithm with an example? (5)

SECTION - D

7. (a) Find the minimum spanning tree of the following weighted graph using Kruskal's algorithm. (5)
(b) Explain the difference between rooted tree and binary tree. (5)
8. (a) Define Group. Prove that the set of integers under addition is a group. (5)
(b) Explain isomorphism and homomorphism with examples. (5)

SECTION - E (Compulsory Questions)

9. Explain the following:
(a) Define tautology and contradiction with example.
(b) Find the power set of $A = \{1, 2, 3\}$?

[P.T.O.]

- (c) Find the dual of the boolean expression : $(A+B)(A+C)$.
- (d) What is Bayes theorem?
- (e) Define simple graph and multigraph.
- (f) Write any two applications of group theory.
- (g) Differentiate between onto and one-one functions?
- (h) What is bipartite graph?
- (i) Define contra positive and converse of an implication?
- (j) What is Hasse diagram? (10×2=20)